

Tail Bounds for Sample Median

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Basics

I was interested in the concentration properties of the sample median, and the extent to which it has better integrability properties than the ‘seed’ random variable.

Let $N \in \mathbf{N}$, $M = 2 \cdot N - 1$ and $X_1, \dots, X_M \stackrel{\text{i.i.d.}}{\sim} p$, where $p \in \mathcal{P}(\mathbf{R})$ has cumulative distribution function F . I want to obtain good estimates on the behaviour of $\nu_N := \text{med}(\{X_i : i \in [M]\}) = X_{(N)}$.

To this end, observe that if $X_1, \dots, X_N \geq t$, then $\nu_N \geq t$; it follows that

$$\mathbf{P}(\nu_N \geq t) \geq F(t)^{M-N} \cdot \bar{F}(t)^N.$$

(Actually, the exact value here is given by $\mathbf{P}(\nu_N \geq t) = \binom{M}{N} \cdot F(t)^{M-N} \cdot \bar{F}(t)^N$, which is bigger by a factor $\gtrsim 2^{2N}$)

If $\bar{F}(t) \in \Omega(t^{-\alpha})$, then this implies that $\mathbf{P}(\nu_N \geq t) \in \Omega(t^{-N \cdot \alpha})$. As such, if p has polynomial tails, then $\text{Law}(\nu_N)$ will generally also have polynomial tails for any fixed N . However, the lightness of those tails rapidly improves with N , to the point that the polynomial aspect quickly becomes exponentially negligible.

Orlicz Integrability of X

Follow some slides of Lugosi, let Ψ be a Young function, and assume the normalisation $\mathbf{E}[\Psi(X)] \leq 1$, so that Markov’s inequality yields that

$$\mathbf{P}(|X| \geq t) \leq \Psi(t)^{-1}$$

for all positive t . As such, for $\delta \in [0, 1]$, it holds with probability $1 - \delta$ that $|X| \leq \Psi^{-1}(\delta^{-1})$. Moreover, the sample median will satisfy $|\nu_N| \leq \Psi^{-1}(\delta^{-1})$ if at least N of the samples lie within this interval, i.e. (assuming that $\delta \leq \frac{1}{2} < \frac{N}{M}$), and we can estimate that

$$\begin{aligned} \mathbf{P}(|\nu_N| > \Psi^{-1}(\delta^{-1})) &\leq \mathbf{P}(\text{Bin}(M, \delta) \geq N) \\ &\leq \exp\left(-\frac{2 \cdot (N - M \cdot \delta)^2}{M}\right), \end{aligned}$$

where the latter inequality follows from Hoeffding’s inequality. (For δ far from $\frac{1}{2}$, Bernstein’s inequality might yield better bounds.)

As such, if t is sufficiently large (in particular, if $\Psi(t) > 2$), then we have the estimate

$$\begin{aligned} \mathbf{P}(|\nu_N| > t) &\leq \exp\left(-\frac{2 \cdot (N - M \cdot \Psi(t)^{-1})^2}{M}\right) \\ &\leq \exp\left(-N \cdot (1 - 2 \cdot \Psi(t)^{-1})^2\right). \end{aligned}$$

Worked Example: Finite Variance

For example, if $\Psi(t) = t^2$, then one ends up with $\mathbf{P}(|\nu_N| > t) \leq \exp\left(-N \cdot (1 - 2 \cdot t^{-2})^2\right)$ for $t \geq 2^{\frac{1}{2}}$. For $\delta > \exp(-N)$, one can thus say that with probability $\geq 1 - \delta$, it holds that

$$|\nu_N| \leq \sqrt{\frac{2}{1 - \sqrt{\frac{1}{N} \cdot \log\left(\frac{1}{\delta}\right)}}}.$$

Using that for $0 \leq v \leq \varphi^{-1} := \frac{\sqrt{5}-1}{2}$, $\frac{1}{\sqrt{1-v}} \leq 1 + v$, one can even say that when $\delta \geq \exp(-\varphi^{-2} \cdot N)$ (note that $\varphi^{-2} \in [0.381, 0.382]$), it holds with probability $\geq 1 - \delta$ that

$$|\nu_N| \leq \sqrt{2} \cdot \left(1 + \sqrt{\frac{1}{N} \cdot \log\left(\frac{1}{\delta}\right)}\right),$$

which might be more usable. In any case, one can ignore an event of probability $\exp(-\Theta(N))$ and obtain a random variable with ‘qualitatively’ sub-Gaussian tails.

Application to Median-of-Means

This behaviour is put to great use in the Median-of-Means estimator. Given $N = A \cdot B \gg 1$ iid samples $\{X_i : 1 \leq i \leq N\}$, one partitions them into $B > 1$ blocks, each of equal size A , and computes the within-block means $\{\mu_j : 1 \leq j \leq B\}$. For initial variables $\{X_i\}$ with some integrability properties (e.g. finite variance, higher-order deviations, etc), the block-wise means $\{\mu_j\}$ will typically have even better quantitative regularity as A grows. For example, if $\mathbf{Var}(X_i) = v$, then it is classical that $\mathbf{Var}(\mu_j) = v/A$.

In what follows, it helps to conceive of the X_i as being centered. Applying the preceding arguments with

$$\nu^\circ = \text{median} \left(\left\{ \sqrt{\frac{A}{v}} \cdot \mu_j : 1 \leq j \leq B \right\} \right), \quad \Psi(t) = t^2,$$

one sees that with probability $\geq 1 - \delta$ (provided that $B \geq \log\left(\frac{1}{\delta}\right)$), it holds that

$$|\nu^\circ| \leq \sqrt{2} \cdot \left(1 + \sqrt{\frac{1}{B} \cdot \log\left(\frac{1}{\delta}\right)}\right),$$

and hence that the actual Median-of-Means estimator satisfies

$$|\nu^{\text{MoM}}| \leq \sqrt{\frac{2 \cdot v}{A}} + \sqrt{\frac{2 \cdot v}{N} \cdot \log\left(\frac{1}{\delta}\right)}.$$

With $B \geq \xi^2 \cdot \log\left(\frac{1}{\delta}\right)$ for some $\xi > 1$, this yields the bound

$$|\nu^{\text{MoM}}| \leq (1 + \xi) \cdot \sqrt{\frac{2 \cdot v}{N} \cdot \log\left(\frac{1}{\delta}\right)}.$$